Jev in a Multi-Step Workflow: Predicting Failures Is Not Knowing Which Reviews Pay

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***Abstract*—Many systems chain an AI model's typed decisions into a workflow and use the model's reported confidence to decide which cases a human should check. We test how well that works, using Jev, TypeSafe's typed decision model and its three primitives (a categorical `choice`, a binary `noul` and an ordinal `score`), on a bug-triage workflow. We ask whether these signals identify erroneous chained decisions, and whether they identify the errors worth human review. We study 442 Mozilla and Linux kernel bugs, then run pre-registered tests on 500 new bugs, four further models and an independent Kubernetes workflow. Failure is disagreement with a target action we construct from reference labels. A frozen decision-feature estimator ranks failures better than final-step confidence (AUROC 0.76 vs. 0.55 on the new bugs), and this holds for all five models. Most of its signal is simply how often the chosen action has been wrong before. The Kubernetes replication, on a chronological split, confirmed this (AUROC 0.78 vs. 0.63), as did three of four further models, including the only other one competent at every step. These error rates do not transfer across label mixes: fitted on a quota-sampled development set, they can rank natural-mix errors below chance. Re-weighting to the deployment mix repairs the ranking, but the label-free estimate of that mix is biased and made ranking worse on Kubernetes. Better failure ranking did not reduce cost. Under the pre-registered costs, no 10% review policy beat no review. An oracle shows that the savings sit in wrongly accepted cases, which the tested signals could not pick out, and whether review pays off depends on how costly a wrong accept is compared with a review.**

***Index Terms*—uncertainty quantification, calibration, structured outputs, decision workflows, selective prediction, escalation, label shift, human review, LLM evaluation, bug triage**

I. Introduction

Typed decision models return answers in a fixed format (a label, a yes/no or a rating) together with a probability or confidence, and such answers can be well formed and still wrong. When several of them are chained into a workflow, a mistake at one step feeds into the next. A common assumption is that the confidence each step reports can be combined into a single, workflow-level signal that says when to hand a case to a human.

Prior work has benchmarked Jev's primitives and calibration on single tasks [1]–[3], and recent work has examined uncertainty across multi-step agent trajectories [4]. We study a different, operational question. *When a workflow chains several typed decisions, which signals identify the cases where it fails, and can those signals identify the specific costly errors for which human review is worthwhile?* A signal that ranks failures well need not help in choosing which of them to review.

A second complication is that Jev's three primitives report uncertainty in structurally different forms: `choice` picks from unordered labels and returns a probability for each plus a scalar confidence, `score` rates an ordered scale and returns an ordinal distribution plus a scalar confidence, and `noul` answers yes/no and returns a single probability with no separate confidence field. Combining these values assumes that they share a scale. We test that assumption.

TypeSafe's documentation describes Jev's `confidence` for `choice` and `score` as a number that "collapses" the output distribution into a value between 0 and 1. It is 1 when all probability is on one outcome and 0 when probability is spread evenly. The documentation calls this "one reasonable way to summarize a distribution, not the only one", and advises setting thresholds from one's own data [5]. For `noul`, it returns "a single probability" with "no separate `confidence`". Our results therefore do not test a calibration claim. Instead, we measure how these documented concentration statistics behave as escalation signals when Jev's primitives are chained, which the documentation leaves open.

Failure here means disagreement between the workflow's action and a target action we construct from reference labels (Section III); we return to targets that do not depend on that construction in Section IX. We ask three research questions. The first (RQ1) is which signals rank workflow failures: the models' confidence, or what the workflow decided. The second (RQ2) is whether those signals stay stable when the label distribution changes between development and deployment. The third (RQ3) is whether signals that rank failures also identify the reviews that reduce cost.

Our contributions are methodological. (i) We measure uncertainty at the level of a whole workflow of structurally different typed decisions joined by an explicit action rule, so that failure is defined on the action the workflow takes rather than on any single answer, and every step-level signal can be evaluated as a predictor of that failure. (ii) We separate two questions that are usually conflated, whether a signal ranks failures and whether acting on it reduces cost, by scoring a fixed review budget against a frozen cost matrix, random review and an oracle bound. (iii) Every confirmatory test was pre-registered and frozen by hash before its data existed, and the main finding was replicated across five models and on a separate, chronologically split workflow. Section III-F gives the details.

For RQ1, a frozen decision-feature estimator ranks failures

better than final-step confidence on 500 new bugs, and the gap replicates across five models from four sources and in an independent Kubernetes workflow (Sections VIII to X). Paired ablations attribute most of the signal to the action error rate, that is, how often the chosen action was wrong on development data. For the two Jev-style models the estimator adds nothing detectable to that rate, and confidence adds little except for the weakest general models.

For RQ2, those error rates are not portable across label mixes: fitted on a quota-sampled development set, they can rank a natural-mix set's errors below chance on targets that bypass our construction. Re-weighting to the deployment label mix repairs the ranking, even with weights estimated from the models' own predictions without test labels when those predictions discriminate the label, but under known synthetic shifts that label-free estimate is biased toward the development mix, and it made ranking worse on Kubernetes.

For RQ3, under the pre-registered costs no 10% review policy beat no review (Section XI). An oracle locates the potential savings in wrongly accepted cases that the tested signals do not isolate, an exploratory sensitivity analysis shows that whether review is worthwhile depends on the ratio of wrong-accept cost to review cost, and a case-level score failed a pre-registered feasibility check, so its confirmatory study was stopped before new data.

Supporting results show that the primitives' confidence signals behave differently inside the workflow, that fixed thresholds set on constructed content do not transfer, and that `score` compresses severity on two trackers, consistent with single-task audits of Jev [1], [2] (Sections IV to VII). The study's own audit process caught seven defects, five of them before the data that would have exposed them existed (Section XII).

We do not claim anything about general calibration, about other Jev versions or other APIs, about domains beyond software issue triage, or about real-world operational benefit. The remaining sections cover related work (Section II) and the methodology (Section III), then the exploratory analyses on the original trackers (Sections IV to VII), the pre-registered prediction tests (Sections VIII to X), and the review-value test (Section XI), followed by the audits and threats to validity (Sections XII and XIII) and the conclusion (Sections XIV and XV).

II. Related work

Evaluations of Jev and open decision models. Jev has been benchmarked zero-shot on 37 datasets (346,009 requests). The authors found good calibration for `choice`, enough to let the model abstain on uncertain items, while its binary probabilities ranked items well but were poorly placed relative to a 0.5 threshold, and thresholds tuned on training data improved F1 [1]. A calibration audit with released per-item outputs (described by its author as exploratory) compared Jev with open and general-purpose classifiers on SST-2 and AG News [2]. It found that calibration changed with the question form (a yes/no form was underconfident, a choice form close to calibrated), that 71.6% of four-class answers sat at exactly 1.00, so a threshold on the top-class probability had only 53 operating points, that one fitted temperature removed most miscalibration, and that no threshold on Jev's scores certified a 5% automatic tier on the four-class task under its procedure; and a comparison on seven political-science replications found its calibration advantage over other models not consistent [3]. An early evidence audit of 28 papers on typed decision models found no independent accuracy advantage yet for the typed readout over comparable label-probability readouts, and proposed an evaluation checklist [6]. Open instruction-tuned models can produce Jev-like decisions by reading answer-label logits in one forward pass [7], the same kind of readout we use for the general models in Section IX. These studies evaluate single requests. None studies `score` inside a workflow, composition of dependent decisions, or the cost of human review. Our results agree with them where they overlap: `choice` confidence is saturated yet discriminative, and thresholds must be validated on the target data, which our transfer result shows directly. On `noul` the measures differ. The audits report calibration and accuracy on sentiment, where the yes/no form was accurate and recalibratable. On our regression question, `noul`'s probability did not discriminate its own correctness (AUROC near or below 0.5). Binary-probability quality looks task-dependent.

Calibration of single decisions. Post-hoc calibration [8], [9] and the miscalibration of modern networks [10] are well studied, and confidence estimation for language models has been surveyed in [11].

Selective prediction and deferral. Risk–coverage trade-offs and reject options [12]–[14], learning to defer [15], [16] and prediction sets in human–AI teams [17] address one decision. Work on human–AI teams shows that training a model for the combined result, including the cost of checking its output, can matter more than its raw accuracy [18]. Closest to our intervention result, Park et al. argue that when only a limited number of answers can be checked, the chance that an answer is wrong and the value of checking it are different quantities: what to review first should also weigh how easily an error can be fixed, how much it matters and what the review costs. On a question-answering benchmark, ranking by this review value left less residual error than ranking by error risk alone [19]. Our intervention finding is an instance of that distinction in a dependent typed-decision workflow, with a frozen cost table and pre-registered tests.

Label shift. When class priors change between training and deployment, posteriors can be re-adjusted to the new priors [20], and the new priors can be estimated from a black-box predictor's confusion matrix and its predictions on unlabeled deployment data, provided that matrix is invertible [21]. Our RQ2 finding is a consequence of this for reliability estimates: an error rate conditioned on a decision is itself a function of the label prior, and we use the second method to correct it without test labels.

Reliability of multi-stage software systems. Component-based

reliability models treat the probability that an error propagates from one component to the next as an architectural attribute that shapes where to place detection and recovery [22]. Bug severity prediction from report text is a long-studied triage task [23]. We study a related propagation problem for chained model decisions, with the escalation point as the recovery mechanism.

Conformal methods and composition. Conformal prediction [24] and conformal risk control [25] give distribution-free guarantees. Composition across stages has been studied for two stages [26] and for multi-stage NLP and LLM pipelines, where errors compound across stages and independently calibrated stages lack a joint coverage guarantee [27]. Error propagation through pipeline steps has also been quantified [28]. Classical reliability theory composes component reliabilities [29].

Cascades and escalation. Cascades route a single query through model tiers. Prior work studies early abstention [30], the optimal depth and deferral thresholds of cascades [31], conformal cascades [32], confidence tuning [33] and judges with escalation guarantees [34].

Combining confidence across components. Fixed combination rules such as the product rule are classical [35]. The product rule's usual justification assumes conditional independence and comparable probability scales, which are the assumptions our strategy S2 makes implicitly. At the system level, a recent taxonomy argues that uncertainty in LLM applications must be studied across the whole compound system, as it propagates within models, across system components and through socio-technical processes [36]. Cost-sensitive deferral to multiple experts under workload limits [37] is closely related to the budgeted escalation setting our follow-up uses.

Ordinal outputs and central tendency. LLM raters have been reported to pull ordinal ratings toward the middle levels, for example when scoring clinical drawing tests, and changing the prompt did not fix it [38]. Few-label calibration of ordinal LLM outputs models such flattening explicitly [39]. Our severity result is consistent with this literature. We observe the same effect in bug triage, under two labeling processes, and find that the primitive's own confidence does not reflect it.

Agents and trajectories. Closest to our workflow framing, a counterexample shows that step-level calibration need not imply trajectory-level calibration for LLM agents, and step errors are empirically coupled along trajectories [4]. Related work calibrates agent trajectories [40] and studies uncertainty propagation in multi-step agentic decisions [41].

Positioning. Prior work benchmarks Jev's primitives on single requests, studies calibration of single decisions, cascades of the same decision and agent trajectories, and argues in general that risk differs from review value. In contrast, we study a dependent workflow of three structurally different typed decisions and, against real labels, held-out data and five models, test which signals predict workflow failure and whether they find the costly errors that make human review worthwhile. We did not find prior work with this exact setup. Our literature search was web-based and did not systematically cover reliability engineering or HCI escalation research (we added the closest work on error propagation, human–AI teaming, deferral and label shift in a later pass). Most of the Jev papers we cite appeared in September and October 2026.

III. Methodology

A. Study architecture

Figure 1 shows the three stages of the study. Bug reports from the Mozilla and Linux kernel Bugzilla trackers, and issues from the Kubernetes GitHub repository, are cleaned, privacy-scrubbed, stripped of text that leaks their labels, and truncated to 2,000 characters. Each report then passes through the same three-step decision workflow, and the workflow's action is compared with a target action computed from the tracker's reference labels, which the model never sees. The comparison defines the failure label on which every signal is evaluated.

B. Workflow, repeats and action rule

All Jev calls go to the TypeSafe API with model `jev-latest`, which resolved to `jev-1.13.0` in every analysed call. Node 1 (`choice`) routes the report to a subsystem: network or storage on the Bugzilla trackers. Node 2 (`noul`) answers a binary question: customer impact on constructed data, regression on real bugs. Node 3 (`score`) rates severity on three levels. Each node reads the evidence plus the upstream decisions, appended by a fixed adapter. Each node is called three times, and the three answers are reduced to one decision: for `choice` and `score` we take the modal decision with a tie tolerance of 0.02, and a node with no two-of-three majority is *repeat-unstable*. For `noul` we threshold the mean probability at 0.5, and its confidence-like value is $2\,|\bar{p} - 0.5|$. The action rule F maps an *unclear* c_2 to *request more evidence*, $c_3 = 2$ to *escalate*, $c_3 = 1$ with $c_2 = 1$ to *defer*, and everything else to *accept* (Table VIII).

C. Target action and failure signals

The target is $a^* = F(c_2^*, c_3^*)$, the same rule applied to the reference labels. F is a construct we defined, not an independently validated outcome, and every failure result is conditional on it. A workflow fails on a report when its action differs from the target, $y = \mathbf{1}[a \neq a^*]$. We compare three kinds of failure signal. The first is the model's own *final-step confidence* (low confidence ranks first), alone or composed across nodes. The second is the *action-rule score* $r(a) = (e(a) + \bar{r})/(n(a) + 1)$, the historical error rate of the action the workflow chose, where $e(a)$ and $n(a)$ count the errors and uses of action a on development data and $\bar{r}$ is the overall development error rate. The third is a frozen logistic estimator on the decisions (E_{dec}) or on the decisions plus the uncertainty signals (E_{all}), fitted on development data only.

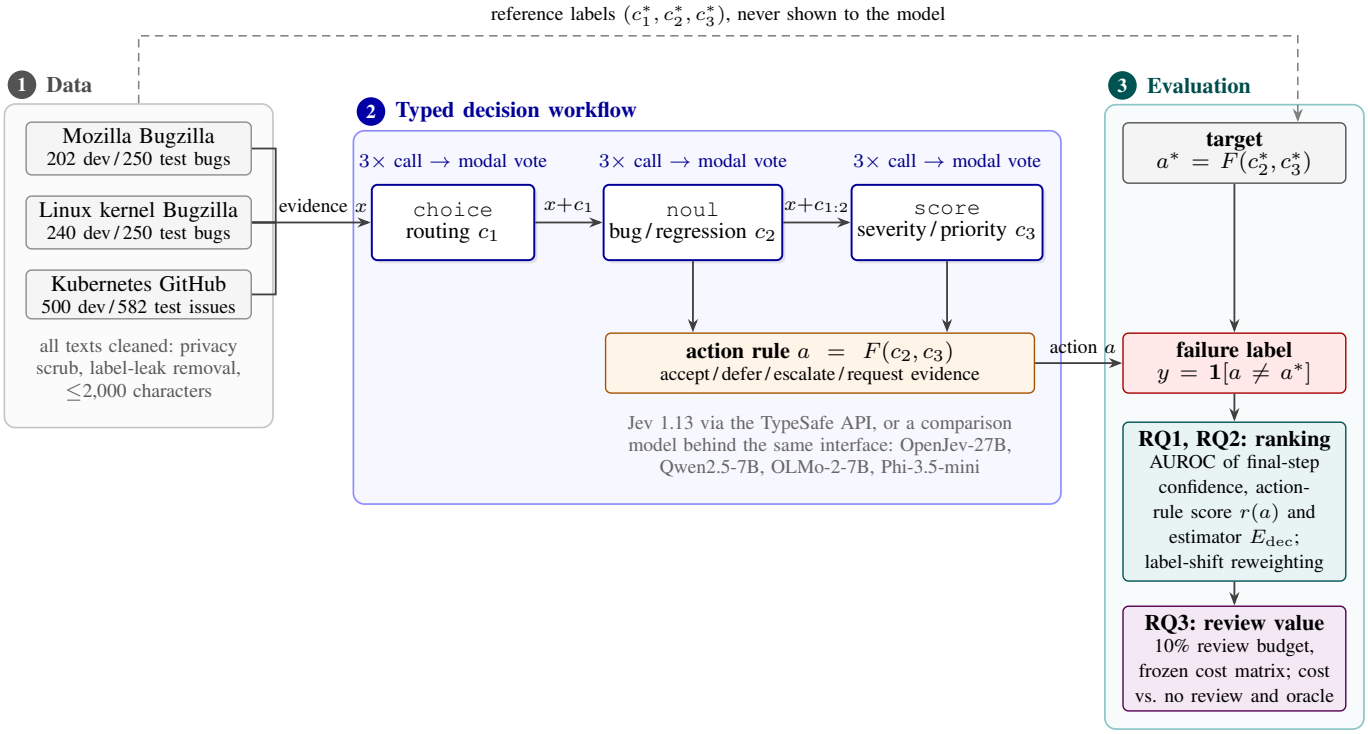


Fig. 1. Study architecture. (1) Bug and issue reports from three public trackers are cleaned and stripped of label leaks. (2) The evidence passes through a dependent chain of three typed decisions, each called three times and reduced by modal vote, with upstream decisions appended to the evidence; an explicit action rule F maps the second and third decisions to an action. The same chain runs on Jev and on four comparison models. (3) Reference labels, never shown to the model, give the target action a^* through the same rule; a failure is $a \neq a^*$. Failure signals are evaluated for ranking (RQ1, RQ2) and for review value under a fixed budget and frozen costs (RQ3).

D. Escalation strategies

The exploratory analyses compare six escalation strategies. S1 applies F to the node decisions, and S4a adds escalation on repeat instability. S2 escalates the $\lceil qn \rceil$ instances with the lowest product of per-node confidences and accepts the rest, where q is the share of non-accept targets, and S5 does the same with per-primitive rank-normalized uncertainty. Strategies 2h and 5h keep S1 and add confidence-ranked escalation of a fixed 10% budget, and R escalates the same budget at random.

E. Evaluation and statistics

Outcomes are $P(Y_{\text{decision}})$, the probability that the action equals a^*, and the cost under a frozen matrix (Table IX), in which a wrong *accept* costs 6–10, any other mismatch 1, and each escalation an extra 1 for human time. Agreement rates carry Wilson 95% intervals. AUROCs use the Mann–Whitney estimator, with 95% percentile bootstrap intervals over bugs ($B = 2{,}000$, fixed seed). Bugs are distinct reports with no shared unit, so we resample bugs. Paired strategy comparisons use the exact McNemar test on discordant pairs, and the difference in $P(Y_{\text{decision}})$ gets a bootstrap interval. Within every resample, the set-relative rankings (S2, 2h) are recomputed. We compare the two trackers through the difference of their paired differences, with independent bootstraps per tracker, and do not compare their p-values. For budgeted overrides, the reference is the exact expected cost of k random overrides. The randomization p-value (10,000 random k-subsets) is conditional on the evaluated bugs, and a bootstrap interval over bugs gives the sampling uncertainty. An interval that covers the null value is read as *no evidence of an effect*, not as evidence of no effect. Confirmatory tests are one-sided paired bootstrap tests with Holm correction within each pre-registered family. The scripts are `paper_statistics.py` and `mozilla_exploratory_analysis.py`.

F. Novelty of the methodology

Table I summarizes the methodological elements of this study and what each one lets us conclude. Together, the pre-registration, the cross-model and cross-workflow replications, and the separation of ranking from value make it unlikely that the negative review-value result is an artifact of chance or of post-hoc analysis choices.

IV. Characterization: Incompatible Uncertainty Semantics

Table II and Fig. 2 show three incompatible profiles. `choice`'s confidence is almost always near 1.0. It still discriminates, because the rare low-confidence answers are where its errors are. The high-confidence error subgroup found on deliberately ambiguous constructed items does not recur on real text (2 of 6 kernel errors are at ≥ 0.90). `noul`'s confidence-like value carries no usable information

TABLE I
METHODOLOGICAL ELEMENTS OF THIS STUDY AND WHAT EACH MAKES IT POSSIBLE TO CONCLUDE.

Element	What we do	What it makes possible
Unit of analysis	A dependent chain of three structurally different typed decisions (categorical, binary, ordinal) mapped to an action by an explicit rule F	Failure is defined on the action taken, so every step-level signal can be compared as a predictor of workflow failure
Failure signals	Final-step and composed confidence, the action-rule score $r(a)$ and frozen estimators, compared by paired tests	Shows whether confidence adds anything beyond what the workflow decided
Prediction vs. value	A fixed 10% review budget scored with a frozen cost matrix against random review, confidence-ranked review and an oracle bound	Separates "ranks failures" from "finds the reviews that reduce cost"
Shift diagnosis	Development and deployment label mixes differ by design; re-weighting with known and label-free (BBSE) mix estimates	Exposes the dependence of decision-conditioned error rates on the label prior
Confirmatory control	Pre-registration, hash-frozen estimators and protocols, dated amendments, independent recomputation	Separates confirmatory from exploratory claims (Table X)
Replication	Five models behind one interface, and an independent, chronologically split Kubernetes workflow	Tests whether the findings belong to Jev or to the workflow design

TABLE II
PER-PRIMITIVE BEHAVIOR OF THE CONFIDENCE SIGNAL. AUROC IS FOR PREDICTING THE NODE'S OWN AGREEMENT WITH THE REFERENCE LABEL, WITH 95% BOOTSTRAP INTERVALS.

	Constructed items	Mozilla (n=202, triage)	Kernel (n=240, reporter)
`choice`	High-confidence error subgroup: mean confidence 0.915, accuracy 0.333 (n=6)	85% at $\geq$ 0.90, agreeing 0.96; AUROC 0.90 [0.83, 0.96]; ECE 0.05	96% at $\geq$ 0.90, agreeing 0.99; AUROC 0.98 [0.95, 1.00]; ECE 0.02
`noul`	24/24 correct, confidence below accuracy in every bin	Agreement flat across bands (0.71–0.84); AUROC 0.53 [0.42, 0.65]; ECE 0.36	Non-monotonic (0.58–0.93); AUROC 0.41 [0.32, 0.50]; ECE 0.36
`score`	Tracks the most severe sentence (6/6); confidence varies by 0.173 with sentence order alone (1 of 3 items)	Confidence $\approx$0.6 at every true level; AUROC 0.58 [0.50, 0.66]	Confidence 0.65 / 0.62 / 0.61 by true level; AUROC 0.57 [0.50, 0.64]

about correctness on either tracker. Its Mozilla interval covers 0.5, and its kernel interval lies at or just below 0.5 (weakly anti-informative). `score`'s confidence does not vary with the true level, so it gives no warning of the systematic bias in Section VI. A product or average of these values combines one discriminative signal (`choice`) with two uninformative ones.

V. FIXED THRESHOLDS LOSE THEIR SELECTIVITY ON REAL TEXT

The two rules in Fig. 3 fail for different reasons. On real text the trigger fired on 85% (Mozilla) and 96% (kernel) of bugs, and the product cut escalated 88% and 78% against targets of 41% and 42% (constructed pilot: 27/30 fired; 73% escalated against 67%). The `choice` trigger is non-selective everywhere, because `choice` confidence is saturated. This design flaw is visible from the trigger's selectivity on input signals alone, before any evaluation. The product cut is distribution-sensitive: it was near its target on constructed data and far off on real data, because real text pulls `noul` toward 0.5, which lowers the product. A fixed threshold is tied to the confidence distribution of its tuning data, and our constructed items did not reproduce the noisier distribution of real text. The two Revision-10 strategies built on these thresholds scored 0.37–0.40 (Mozilla) and 0.34–0.37 (kernel). We replaced them with rank-based rules at a matched escalation rate before any evaluation.

VI. REAL-DATA VALIDATION ON TWO TRACKERS

Data. Mozilla: 202 fixed Firefox networking and storage bugs. Severity S1–S4 is assigned by triage and mapped to levels 2/2/1/0, and the regression flag is a keyword. Bugs whose severity was revised were excluded. *Linux kernel:* 240 fixed networking and storage bugs. Severity (blocking/high/normal/low $\rightarrow$ 2/2/1/0) is set by the reporter, and the regression flag is a dedicated field. Severity is rarely changed after filing: 15 of the 240 sampled bugs, all moving away from the default "normal". Both samples are balanced at 40 per (subsystem, level) cell where available. Texts are cleaned,

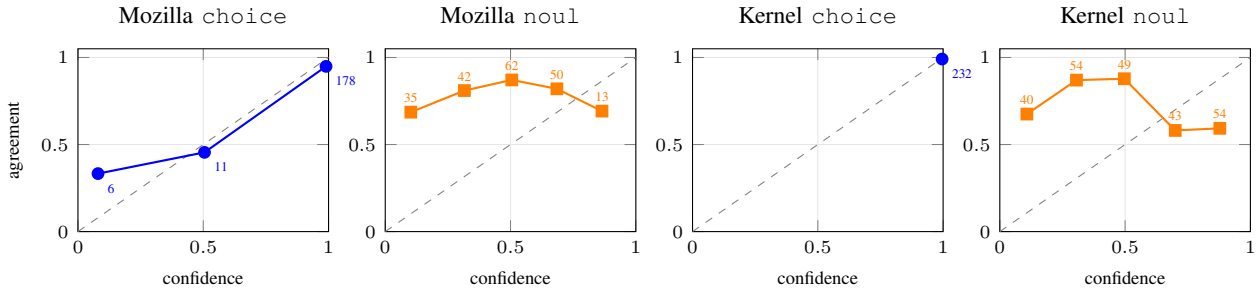


Fig. 2. Reliability diagrams (five equal-width confidence bins, bins with $n \geq 5$; the label is n; dashed line: perfect calibration). `choice` puts nearly all bugs in its top bin, and that bin is accurate. `noul`'s confidence-like value $2|\bar{p} - 0.5|$ sits far below its agreement (underconfidence), and agreement does not rise with it.



Fig. 3. Selectivity of the two fixed-threshold rules, set on constructed data and measured on each rule's own input signals. The trigger fires on 85–96% of instances everywhere. The product cut is near its target on the constructed pilot and far above it on real text.

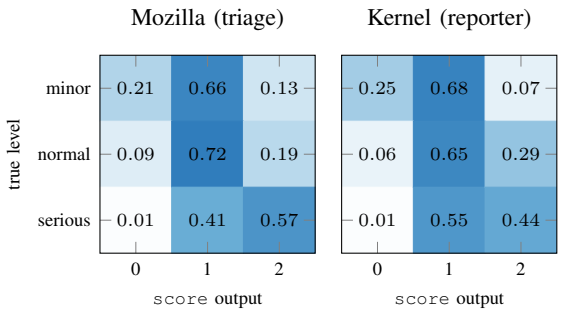


Fig. 4. Row-normalized confusion of `score` against reference severity (repeat-unstable outputs excluded). On both trackers the middle column absorbs mass from both extremes: minor bugs are rated up, serious bugs down.

privacy-scrubbed and truncated to 2,000 characters. Agreement is with the reference label, not ground truth.

Routing and regression. `choice` agrees with the component label well above baseline on both trackers (Table III). It still agrees 0.84 on Mozilla texts with no subsystem keyword. `noul`'s errors change direction between trackers. On Mozilla it mostly misses regressions (23 misses vs. 17 false alarms). On the kernel it over-flags them (63 false alarms vs. 3 misses), which leaves it below the always-"no" baseline. The kernel false-alarm rate is the same on bugs explicitly marked "not a regression" (33/100) as on bugs with the field unset (30/94). So unset labels do not explain it, and on bugs with a set field `noul` beats its baseline (110/146 vs. 100/146).

Severity compression. Figure 4 shows the row-normalized confusion of `score`. On both trackers, most minor bugs are rated higher than their label: 79% [0.67, 0.88] on Mozilla and 75% [0.65, 0.83] on the kernel. A large share of serious bugs are rated lower: 43% and 56%. The middle level absorbs both. With Mozilla's own severity definitions in place of ours, serious bugs improve but normal bugs are pushed up, and agreement moves only from 0.53 to 0.56, so the compression is not caused by our severity mapping. The two trackers differ in project, bug population and labeling process. They share Jev, the prompts, our sampling design and, plausibly, training-data exposure, so the replication argues against an artifact of one labeling process, although shared factors such as the prompts may contribute. The pattern matches central-tendency bias reported for LLM raters in other domains [38].

VII. Composition (Exploratory)

All analyses in this section were specified after the Mozilla results had been seen. They are hypothesis-generating. The pre-registered tests that followed are reported in Sections VIII, IX and XI.

The naive comparison is confounded. On Mozilla, decision composition (S1) beats confidence composition (S2) by 0.17 [0.08, 0.27] in $P(Y_{\text{decision}})$, with 61 vs. 27 discordant pairs (exact McNemar $p = 0.0004$). But S2 never reads the node decisions, while a^* escalates on the severity decision. To be right on a serious bug, S2 needs low confidence to mean high severity, so the comparison mostly measures whether confidence can stand in for severity. On the kernel the difference is smaller and inconclusive: 0.06 $[-0.03, 0.15]$, 66 vs. 51 discordant pairs, $p = 0.20$. The difference between trackers is 0.11 $[-0.01, 0.24]$. Its interval covers zero, so we have no evidence that the effect differs between trackers, only a weaker estimate on one of them.

The fair question, and two different targets. Does confidence add anything beyond the decisions? Starting from the decision-based action F, we also escalate the k least-confident bugs that F did not escalate, for every k, and compare with the

TABLE III
NODE AGREEMENT WITH REFERENCE LABELS (WILSON INTERVALS). EACH BASELINE IS THE MOST FREQUENT LABEL.

	Mozilla		Kernel	
Node	Agreement [95% CI]	Baseline	Agreement [95% CI]	Baseline
`choice`	0.89 [0.83, 0.92]	0.59	0.975 [0.95, 0.99]	0.50
`noul`	0.80 [0.74, 0.85]	0.74	0.725 [0.67, 0.78]	0.81
`score`	0.53 [0.46, 0.60]	0.40	0.44 [0.38, 0.50]	0.33

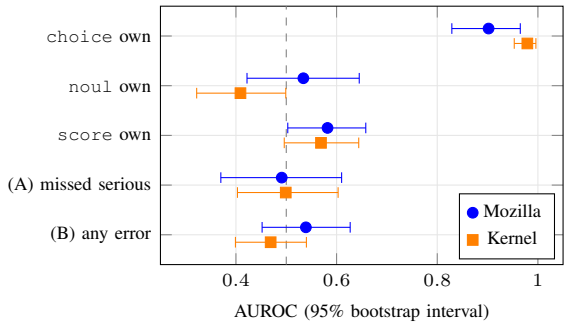


Fig. 5. Every AUROC in the paper with its interval; dashed line: chance. Only `choice`'s confidence discriminates its own errors. Composed confidence is at chance for both escalation targets, (A) and (B).

exact expectation under random selection (Fig. 6). We report two AUROCs with different prediction targets. Target (A) is *missed serious bugs*: among bugs that F did not escalate, does low composed confidence rank those whose target is *escalate* first? This is what an override budget can fix, and the AUROC is 0.49 [0.37, 0.61] on Mozilla and 0.50 [0.40, 0.60] on the kernel. Target (B) is *any decision error*: across all bugs, does low composed confidence rank those where F's action is wrong first? The AUROC is 0.54 [0.45, 0.63] on Mozilla and 0.47 [0.40, 0.54] on the kernel. Every interval covers 0.5 (Fig. 5). The data show no evidence that composed confidence ranks either target better than chance, but they cannot rule out a modest signal (AUROC up to about 0.6).

Budgeted overrides, scored by cost. At a fixed budget of $k = \lceil 0.1n \rceil$ overrides ($k = 21$ on Mozilla, 24 on the kernel), let T be the total cost of the confidence-ranked overrides and μ the exact expected total under k overrides drawn uniformly from the same candidates. We run two analyses. A randomization test, conditional on the evaluated bugs, asks whether the ranking selected worse or better overrides than random selection on these bugs. Its statistic is $|T - \mu|$, and its two-sided null distribution comes from 10,000 random k-subsets of the candidates. A bootstrap over bugs asks how stable the per-bug difference $(T - \mu)/n$ is under resampling of the bugs, with rankings and μ recomputed in each resample. On Mozilla, $T = 390$ against $\mu = 357.2$. The observed value sits at the edge of the null distribution (2.5–97.5% range 323–388), so $p = 0.05$. The per-bug difference is $+0.16$ [-0.01, 0.24], and 97% of resamples are positive. That is borderline evidence that the product ranking chose worse overrides than random on Mozilla. On the kernel, $T = 338$ against $\mu = 358.6$ ($p = 0.18$), a per-bug difference of -0.09 [-0.23, 0.06]: inconclusive, and in the opposite direction. These analyses are exploratory and uncorrected for multiplicity. We read them as no consistent benefit from confidence-ranked overrides, with one borderline harmful estimate that did not replicate, and not as an established harm.

A structural reason. In this workflow the only discriminative confidence (`choice`, AUROC 0.90–0.98) belongs to the one node that F does not use. The two nodes that determine the action have uninformative confidence.

VIII. A PRE-REGISTERED HELD-OUT TEST

Design. The exploratory analyses above suggested that a model's decisions carry the information its confidence lacks. We tested this prospectively. We trained one decision-feature estimator (E_{dec}: L_2-regularized logistic regression on the `score` level, the `noul` decision and the workflow action, with $\lambda = 1$) on all 442 previously collected bugs and froze it by hash before any new data existed. We then collected 500 new bugs, 250 per tracker, sampled at each tracker's natural mix with no overlap with the training bugs, which required 4,500 Jev calls. Two hypotheses were pre-registered: E_{dec} ranks wrong workflow actions better than low final-step confidence (HR1), and E_{dec} ranks above chance on each tracker (HR2), both by one-sided bootstrap tests with Holm correction.

Result. Both hypotheses are supported. For HR1, E_{dec} reaches AUROC 0.763 against 0.549 for final-step confidence, a difference of $+0.214$ (one-sided 95% lower bound $+0.158$). For HR2, the AUROC is 0.811 on Mozilla (lower bound 0.762) and 0.697 on the kernel (lower bound 0.648). Per tracker, the difference is $+0.279$ [$+0.191$, $+0.374$] on Mozilla and $+0.124$ [$+0.023$, $+0.225$] on the kernel, weaker but excluding zero.

The result survives three integrity checks. Tracker links (duplicates, see-also, blocks, depends-on, regressions) connect 9 held-out bugs to development bugs. Without them E_{dec} reaches 0.770 against 0.551. On the 70 Mozilla bugs filed in the 12 months before sampling, E_{dec} reaches 0.847 against 0.519. Refitting the estimator after removing every development bug linked to a held-out bug gives 0.772 against 0.551 (difference $+0.221$ [$+0.154$, $+0.288$]). A separate script that shares no code with the analysis rebuilds the reduction, action rule, features, AUROC and bootstrap from the raw responses and reproduces every point estimate exactly. The estimator's

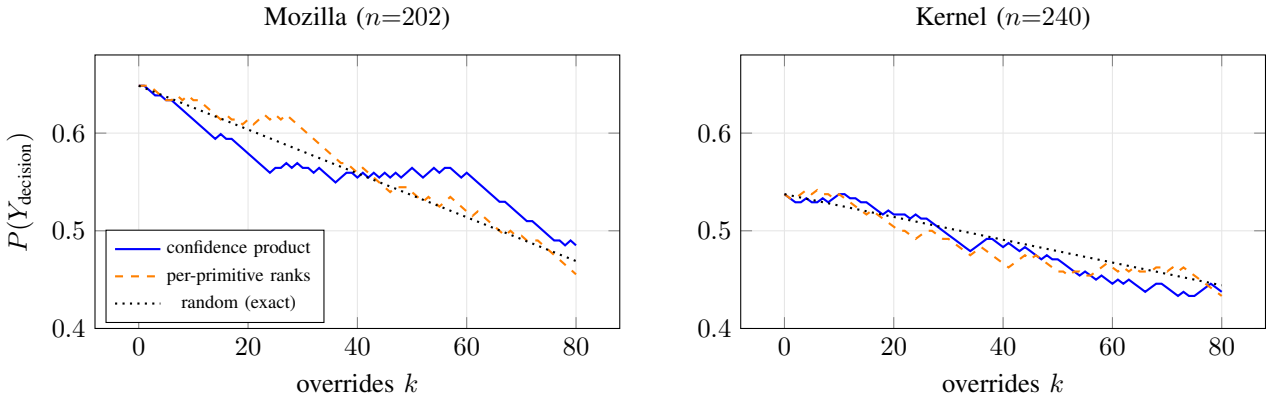


Fig. 6. Adding confidence-ranked escalations on top of decision composition. At $k = 0$ the curve is S1. Neither ranking consistently improves on random selection of the same number of cases: deviations are a few points at most and change sign between rankings and trackers. For example, per-primitive ranks run slightly above random on Mozilla for $k \approx 15$–35, and the confidence product runs below.

predicted failure rate matches the observed rate overall, but it outputs only 7–8 distinct risk levels (ECE 0.18–0.20).

IX. Cross-model replication

Models. To test whether the result is specific to Jev, we ran the same 942 bugs, questions and three-step adapter through four further models. OpenJev 27B [42] is an independently trained Jev-style decision model (Hugging Face `openjev/openjev`, commit `b4a89f5`; several unrelated projects share the name), served on one H100 exactly as its authors specify, with their calibrated readout. Its 2,826 requests had 0 failures. (An unofficial architectural reconstruction of Jev also exists [43], but it has no trained weights, so it could not be evaluated.) The other three are general instruct models, Qwen2.5-7B [44], OLMo-2-7B [45] and Phi-3.5-mini [46], each run behind a Jev-compatible readout: next-token probabilities over the allowed labels (as in [7]), with TypeSafe's documented confidence formulas. Each model's estimator was trained on that model's own outputs for the 442 training bugs and tested on its 500 held-out bugs. The pre-registered tests mirror HR1 and HR2, with Holm correction across all 8 tests.

Why these models. Jev is itself a language-model-based decision service used zero-shot: it receives text and a typed question, with no training on the task. The comparison that isolates Jev is other models used the same way, through the same questions, adapter and confidence formulas. We chose open-weight models at the small end of zero-shot classification, 3.8B to 7B parameters, the scale chosen when routine classification must be cheap, from three independent families with different training data, plus OpenJev, the only openly available model trained for Jev's interface (27B). Open weights make every run reproducible. Our aim is to test whether the failure-prediction findings are specific to Jev or hold for any model that makes typed decisions through this interface, not to rank models by accuracy. Weaker models help here because they yield more failures to predict. A supervised classifier trained on each workflow's labels, such as a linear model on bag-of-words

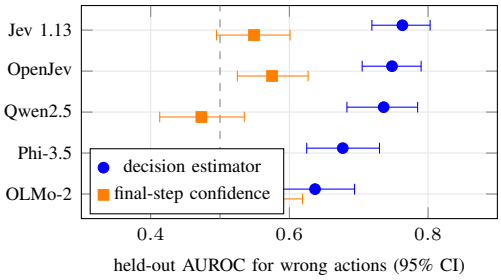


Fig. 7. Across five models, the decision-feature estimator ranks wrong workflow actions better than the model's own final-step confidence. The intervals overlap only for OLMo-2.

features, is the simpler alternative wherever such labels exist, but it answers a different question and is not evaluated here.

Deviations, disclosed. We recorded three pre-registration amendments. OpenJev was added before any of its outputs existed; OLMo-2 needed a newer `transformers` release; and a readout bug, in which a SentencePiece word-start token was shared by all digit labels, was found after interim results had been seen, so all three general models were re-run under one corrected readout. The conclusions did not change.

Result. All 8 pre-registered tests are supported (Fig. 7, Table IV). The decision estimators reach AUROC 0.64–0.76 and final-step confidence 0.40–0.58. The weakest model is OLMo-2 (0.637 vs. 0.551; one-sided $p = 0.037$, passing at the last Holm step). Jev's frozen estimator, never refitted, reaches 0.76 [0.72, 0.80] on OpenJev's outputs, as well as OpenJev's own estimator does.

Most of the signal is the action error rate. A class-frequency rule, "the error rate of the action the workflow chose" (estimated on training bugs only), matches the estimator for Jev (0.76) and OpenJev (0.75), and comes close for the others (0.58–0.70). Single uncertainty signals each sit at chance for workflow failure: routing confidence at 0.48–0.52, regression confidence at 0.45–0.56 and severity entropy at 0.40–0.55. This is partly structural: the action and the target are both

TABLE IV

PER-MODEL RESULTS ON THE SAME 500 HELD-OUT BUGS. AUROC COLUMNS RANK WRONG WORKFLOW ACTIONS (95% BOOTSTRAP CI): THE DECISION-FEATURE ESTIMATOR FITTED ON EACH MODEL'S OWN TRAINING OUTPUTS, THE MODEL'S FINAL-STEP CONFIDENCE, THE ACTION-CONDITIONAL ERROR-RATE RULE, AND JEV'S FROZEN ESTIMATOR APPLIED UNCHANGED. COST COLUMNS ARE FROM THE PRE-REGISTERED INTERVENTION TEST (SECTION XI); THE ORACLE IS AN UPPER BOUND THAT MUST SPEND ALL 50 REVIEWS, AND "WRONG ACCEPTS" IS THEIR SHARE OF NO-REVIEW COST. ABSOLUTE FAILURE RATES ARE NOT A CLEAN QUALITY COMPARISON, BECAUSE THE JEV-STYLE AND GENERAL MODELS USE DIFFERENT INTERFACES.

			Held-out AUROC for wrong actions				Cost at 10% review budget			
Model	Type	Fail rate	Decision est.	Final-step conf.	Action rule	Jev est.	No review	Best review arm	Oracle	Wrong accepts
Jev	TypeSafe, hosted	0.43	0.76 [0.72, 0.80]	0.55 [0.49, 0.60]	0.76	—	658	681 (conf. product)	370	52%
OpenJev	Jev-style, 27B	0.47	0.75 [0.70, 0.79]	0.57 [0.53, 0.63]	0.75	0.76	660	674 (confidence)	441	40%
Qwen2.5	general, 7B	0.74	0.74 [0.68, 0.79]	0.47 [0.41, 0.54]	0.70	0.74	903	910 (estimator)	856	14%
Phi-3.5	general, 3.8B	0.72	0.68 [0.62, 0.73]	0.40 [0.34, 0.46]	0.69	0.70	831	890 (confidence)	890	4%
OLMo-2	general, 7B	0.80	0.64 [0.58, 0.69]	0.55 [0.48, 0.62]	0.58	0.64	893	916 (all arms tie)	916	1%

TABLE V

EXPLORATORY: HELD-OUT AUROC OF THE DECISION-ONLY ESTIMATOR ON TWO TARGETS THAT DO NOT PASS THROUGH THE ACTION RULE F, FITTED ON THE QUOTA-SAMPLED DEVELOPMENT BUGS ("DEV FIT") OR ON THE SAME BUGS RE-WEIGHTED TO THE NATURAL LABEL MIX, AGAINST FINAL-STEP CONFIDENCE. BOLD: RE-WEIGHTED ESTIMATOR ABOVE CONFIDENCE.

	Severity wrong			Any node wrong		
Model	dev fit	re-weighted	conf.	dev fit	re-weighted	conf.
Jev	0.28	**0.79**	0.61	0.56	**0.76**	0.56
OpenJev	0.28	**0.79**	0.63	0.62	**0.81**	0.56
Qwen2.5	0.62	**0.74**	0.46	0.76	**0.73**	0.40
Phi-3.5	0.31	**0.67**	0.41	0.67	**0.77**	0.29
OLMo-2	0.50	**0.62**	0.57	0.42	0.42	0.49

outputs of the same rule F, so an action's error rate is a property of the workflow as much as of the model. Paired bootstrap contrasts quantify the remaining contributions. For Jev and OpenJev the decision estimator adds nothing detectable beyond the action rule ($+0.000$ [-0.012, $+0.011$] and $+0.002$ [-0.007, $+0.011$]); for Qwen2.5 and OLMo-2 it adds a little ($+0.035$ and $+0.057$, intervals excluding 0). Adding the uncertainty features to the decision features raises held-out AUROC by +0.006 (Jev), +0.004 (OpenJev) and +0.012 (Qwen2.5), none distinguishable from 0, but by +0.054 (Phi-3.5) and +0.078 (OLMo-2), both with intervals excluding 0. For the Jev-style models confidence is redundant with the decisions. For the weaker general models raw token probabilities carry some extra information.

Beyond the constructed target (exploratory). Because the target a^* is a construct, we also asked whether decisions predict failures on two targets that bypass F: the severity answer being wrong, and any node being wrong (Table V). Fitted on the development bugs, several decision estimators rank held-out severity errors far below chance (0.28 for Jev and OpenJev). The cause is label-prior shift: a decision's error rate depends on the label mix. In the quota-sampled development set Jev's level-1 answers are 60% wrong. In the natural-mix held-out set, where most bugs are level 1, they are 20% wrong. Re-weighting the development bugs to the natural label mix repairs this: the decision estimator then beats final-step confidence on both targets for four of five models (Jev 0.79 vs. 0.61 on severity), the exception being OLMo-2 on any node. These oracle weights use the held-out sample's label counts. A deployment-feasible version uses no test labels: black-box shift estimation [21] infers the deployment severity mix from the development confusion matrix of each model's `score` answers and the distribution of its answers on the new bugs. It recovers almost all of the oracle gain for Jev (0.74 on severity, 0.78 on any node), OpenJev (0.78, 0.80) and Qwen2.5 (0.74 on severity), and none for Phi-3.5 and OLMo-2, whose severity answers are nearly all "blocking", so their confusion matrices are near-singular and their answers carry no information about the label mix. So decision-conditioned error rates are informative beyond the constructed target, but only when estimated under the deployment label distribution, which can be estimated without labels only for models whose answers discriminate the label.

This label-free repair has an important caveat: BBSE improves the ranking without recovering the deployment mix. Under known synthetic shifts, its estimate is biased toward the development mix (for Jev it underestimates a 50% share of serious bugs by 0.21). On the held-out bugs the estimate collapses onto the middle severity level (for the kernel, $[0, 1, 0]$ against a true $[0.02, 0.73, 0.25]$). The re-weighted fit is then dominated by middle-level bugs, as the natural-mix set is, so the decision-to-error relationship takes the right direction even though the other levels are effectively dropped. Where the estimate points the wrong way, the repair fails: for Phi-3.5 it puts no weight on the middle level, and ranking does not improve. The gains for Jev, OpenJev and Qwen2.5 may therefore depend on the middle level being dominant. Section X tests this on a workflow where it is not.

The constructed-target result itself does not depend on how labels are mapped: with Mozilla's S2 counted as level 1, or kernel bugs with default severity or unset regression excluded, decisions still beat confidence for every model, except OLMo-2 when default-severity kernel bugs are removed.

Failure modes differ by model family. The two Jev-style models, Jev and OpenJev, both pull severity to the middle level. The general instruct models push severity to "blocking" for 67–100% of bugs, whatever the true level, and are wrong far more often (72–80% vs. 43–47%). Because the two groups use different interfaces, the absolute failure rates are not

a clean quality comparison, but the within-model contrast between the estimator and confidence is.

X. An independent workflow: Kubernetes issue triage

Setup. To test whether the predictive finding is a property of our bug-triage setup, we pre-registered a replication in an independent workflow before any of its data existed: issue triage in the Kubernetes GitHub repository. It differs in platform, labelling process (triager-applied priority and SIG labels, reporter-chosen bug/feature forms), three-way routing (network, storage, node), the binary question (bug or not) and the ordinal question (priority). The split is chronological: development on 500 issues from 2015–2019, test on the 582 latest eligible issues from 2021–2026, with linked issues removed across the split. Leakage of labels into the text (bot commands, template headings, team mentions) was removed and audited, and `jev-1.13.0` was pinned. All of the protocol's seven amendments, including a one-time widening of the test window after the first feasibility check failed, were written before any model output existed. Development and test differ at once in period, issue-form use and label prevalence, so this is a combined test of temporal and distribution shift.

Result. The primary hypothesis is supported: the action error rate ranks test-set failures better than final-step confidence, 0.776 against 0.628, a paired difference of $+0.148$ [$+0.091$, $+0.207$], above chance as well. As in the original study, the signal is mostly the action: the decision estimator adds a little ($+0.004$ to $+0.017$) and confidence adds nothing. The contrast holds for the any-node target, within SIG Node, and on issues triaged within 30 days.

Four further models. A second pre-registration, written before any of their outputs existed, ran the same workflow and analysis on the four comparison models (1,082 issues each, no failures), with Holm correction over all eight primary tests. Results differed by whether a model was competent at every step. Only the two models trained for typed decisions beat chance on every step (routing, bug or not, and priority): Jev and OpenJev-27B, whose priority accuracies were 0.371 and 0.404 against a chance level of one third. For both, the action rule beat final-step confidence (OpenJev-27B: 0.796 vs. 0.569). The three general instruct models handled routing and the bug question but not priority (accuracy 5–27%), so their workflows were wrong on most issues (wrong-action rates up to 95%). This competence check was added after the results were seen, applied to every model, and does not replace the primary result. Among these weak models the rule still beat confidence for Qwen2.5 (0.710 vs. 0.555) and OLMo-2 (0.850 vs. 0.255), but largely by flagging the one action each model chose almost always. It was not replicated for Phi-3.5 (0.519 vs. 0.674), which escalated 580 of the 582 test issues and so left the rule nothing to rank. By the pre-registered rule (at least three of four) the finding holds on Kubernetes.

The label-free correction did not transfer. The pre-registered label-shift test is inconclusive by its frozen rule, with a negative estimate: re-weighting a deliberately mismatched fit by BBSE lowered its AUROC from 0.699 to 0.348 (-0.350 [-0.428, -0.271]). The mechanism is the one identified above. BBSE was numerically stable (condition number 39) but again estimated the deployment mix as entirely middle level, $[0, 1, 0]$, whereas the true mix was $[0.69, 0.26, 0.05]$, dominated by the lowest level, so the weighting pushed the fit in the wrong direction. This supports the explanation in Section IX that the earlier BBSE gains relied on a dominant middle level.

XI. Does the predictive signal reduce cost?

Design. We pre-registered the test before computing any cost outcome. Each model gets a budget of 50 reviews (10%) on the held-out bugs. A review escalates the case to a human and is priced with the frozen cost matrix. The primary arm (A1) ranks candidates by the *action-level expected saving* of a review, estimated on training bugs. It is cost-aware rather than error-rate-based, because a wrong *defer* costs no more than an escalation. The comparators are confidence-ranked review (B) and random review (C).

What we found. Two of ten pre-registered tests are supported: A1 beats random for Jev and OpenJev, by about 10–12 cost units. A1 is not shown to beat confidence for any model (for Jev its point estimate is 4 units lower, with an interval spanning zero). Under the frozen table, every review arm costs more than no review at all. For Jev, 658 with no review against 681–714 with review (Fig. 8).

Sensitivity to costs (exploratory). The pre-registered result holds under one frozen cost table. In a post-hoc analysis planned before computing it, we re-scored the same frozen rankings on a grid of cost ratios within a factor of 2 of that table (Table VI). The conclusion is cost-robust only for OLMo-2. For Jev, OpenJev and Qwen2.5, review reduces cost when wrong accepts cost twice as much (for Jev, 44–69 units below no review) or when review costs half as much, because the break-even precision p^* then falls to about the 14–16% share of wrongly accepted cases. Some reversals are small (Jev -2 at $m = 1$, $h = 0.5$) and lie well inside the intervals of the original tests. The ranking of arms is more stable than their sign relative to no review: the action rule beats random for both Jev-style models in 12 of 16 cells. Imperfect reviewers shrink but do not reverse these savings: if a review resolves correctly only 80% of the time, Jev's saving at twice the wrong-accept cost falls from 44 to 26 units. The stopping decision below was made under the frozen table and, by its rule, is not reopened.

Where the cost is. A descriptive follow-up separates two regimes. For the Jev-style models, 40–52% of no-review cost comes from wrongly accepted serious or regression bugs, and an oracle that picked the right 50 cases would save 33–44%. But those cases are 15–16% of accepted cases. The action rule cannot separate them, because every accept looks alike to it, and confidence cannot either: among Jev's 245 accepted held-out cases, the decision estimator, final-step confidence and `score`'s probability of the top level rank the 38 wrong

TABLE VI

EXPLORATORY COST SENSITIVITY: BEST FEASIBLE ARM MINUS NO REVIEW (NEGATIVE, IN BOLD: REVIEW REDUCES COST), WITH RANKINGS FROZEN AS PRE-REGISTERED. m SCALES WRONG-ACCEPT COSTS, h IS THE HUMAN COST PER ESCALATION, AND p^* IS THE BREAK-EVEN PRECISION FOR REVIEWING AN ACCEPTED CASE WHOSE TARGET IS *escalate*. †THE PRE-REGISTERED TABLE. POINT ESTIMATES ONLY.

m	h	p^*	Jev	OpenJev	Qwen2.5	Phi-3.5	OLMo-2
0.5	0.5	25%	+29	+25	+21	+52	+20
0.5	1	33%	+54	+50	+46	+77	+28
0.5	2	50%	+104	+100	+96	+127	+45
1	0.5	14%	**-2**	**-11**	**-18**	+34	+14
1	1†	18%	+23	+14	+7	+59	+23
1	2	27%	+73	+64	+57	+109	+40
2	0.5	7%	**-69**	**-83**	**-97**	**-2**	+4
2	1	10%	**-44**	**-58**	**-72**	+23	+13
2	2	14%	+6	**-8**	**-22**	+73	+30

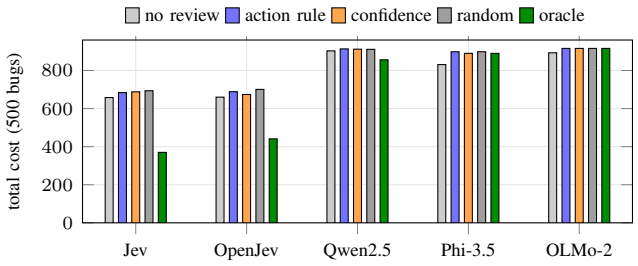


Fig. 8. Total cost at a 10% review budget under the pre-registered cost table. No feasible strategy beats no review. A perfect reviewer (oracle, an upper bound using held-out outcomes) would cut Jev's cost by 44% and OpenJev's by 33%. For the general models it saves little or nothing: they escalate most cases themselves, leaving few reviewable errors (Phi-3.5's oracle exceeds no review because 50 reviews must be spent on 63 candidates, only 4 of which gain).

accepts at AUROC 0.46, 0.49 and 0.49, with 95% intervals that all include chance (for example 0.49 [0.37, 0.60] for confidence). With so few positives this shows that the tested signals did not establish useful discrimination for Jev, not that none is possible. For OpenJev, `score`'s top-level probability does carry a weak within-action signal (0.68 [0.56, 0.78]), and a paired comparison puts it above confidence by $+0.14$ (95% interval $[+0.04, +0.22]$). For Jev, none of the tested signals isolated these cases. For the general models, cost is dominated by over-escalation (*escalate* when the target is *accept*), which more review cannot reduce, and even the oracle saves at most 5%. In both regimes, action-level error rates point to the most frequent errors, whereas cost is concentrated in rarer wrong accepts.

A case-level score, and a stopping decision. We then asked whether a per-case score could find the expensive errors. It was developed on the 442 training bugs only, under a plan committed before any score was computed. A parameter-free plug-in score was selected by a pre-stated rule over a small logistic model and frozen. It ranks accepted cases by expected saving from review, $9p_2 + 6p_1q - 2(1 - p_2 - p_1q)$, using Jev's `score` mass on levels 1 and 2 (p_1, p_2) and its `noul` probability q. On the training mix it cut cost by 12% against no review (630 vs. 713) and captured 22% of the oracle saving. But the training bugs were quota-sampled. Reweighted to each tracker's natural label mix, a 10% budget is projected to cost more than no review, with power near zero. Before generating any fresh case, we pre-registered a feasibility check that varied only the review budget (1–10%). Viability required power ≥ 0.80 to show lower cost than no review, with at most 1,000 fresh bugs, and a negative median projected cost change. Each simulated study first resampled the training bugs, so uncertainty in error prevalence and ranking quality was propagated. No budget was viable. The highest power was 0.20 (1% budget, 1,000 bugs), with a median change of -2 and an 80% range of $[-46, +20]$, and every budget of 2% or more had a positive median cost change and power below 0.15. By the binding stop rule, the confirmatory study was not run, no further designs were searched, and no fresh API calls were made. A failed feasibility check does not show that review cannot help: the oracle bound implies substantial savings for a sufficiently precise selector. The signals we tested were not precise enough to identify those cases under the pre-registered costs.

XII. LESSONS FROM THE STUDY'S OWN AUDITS

Table VII lists the defects the study's own audits caught. Five of seven were found before the data that would have exposed them existed. The exceptions are the confound, found after a result, and the bootstrap seed, found by re-running every analysis from a clean checkout. It moved two lower bounds by at most 0.005 and changed no test outcome. The confound would have been caught earlier had we checked, for every strategy and every target class, whether the strategy had access to the information needed to be correct. We suggest running that check, along with measuring each trigger's selectivity on pilot inputs, before freezing an escalation policy.

XIII. THREATS TO VALIDITY

Construct validity. The failure label is disagreement with a constructed target, $a^* = F(\cdot)$, and all strategy results are conditional on it. Its inputs are tracker labels, which are not truth. Mozilla's triage sample excludes re-triaged bugs, which biases agreement upward. Kernel labels are reporters' claims, and the kernel's middle level is a default value. An exploratory pilot in which one AI assistant chose the right action for 50 held-out bugs, blind to labels and outputs, agreed with the constructed target at about chance level (Cohen's κ between -0.157 and $+0.015$ under two pre-stated readings of its categories). With a single rater and no inter-rater baseline, this cannot tell whether the target or the judge is at fault. It does make the limitation concrete. Every failure result in this paper is disagreement with an operational target built from tracker labels, and we have not shown that this target matches what a competent evaluator would choose. A pre-registered two-human validation is designed and not yet run. The review-value analysis uses one budget and one cost table, both frozen in advance. Real operational costs are unknown, reviewers

TABLE VII
DEFECTS FOUND BY THE STUDY'S OWN AUDITS.

Defect	Detected by	When
A "primitive-aware" arm structurally identical to the baseline	inspecting the action rule	before evaluation
A trigger firing on ≈90% of inputs	selectivity on input signals only	before evaluation
A threshold sensitive to the data distribution	smoke test vs. pilot	before the full run
Four silent runner defects (e.g. the literal "None" shown to the model; partial data analyzed)	adversarial probing	before the live run
The confidence-vs-decisions confound	post-hoc review	after the Mozilla result
Template content whose labels were recoverable from punctuation alone (100%) and whose sentences leaked cross-label cues	independent review (separate AI agent)	before the controlled study
A bootstrap seed derived from Python's salted string hash, making two reported lower bounds vary between runs	clean-checkout re-run of all analyses	after publication of the bounds

are assumed correct, and the oracle is an upper bound, not a strategy.

Internal validity. The 500 held-out bugs were fresh for the prediction tests only. The intervention test was pre-registered after their prediction results were known (no cost outcome had been computed), and the cost-sensitivity analysis is post hoc on the same bugs, so neither is an independent confirmation. Decision-conditioned error estimators fitted on the quota-sampled development bugs can invert on a natural-mix set (Table V). The label-free correction (BBSE) corrects only the severity dimension and assumes pure label shift. It cannot work for models whose answers do not discriminate the label, and these analyses are exploratory and use the same held-out bugs. The constructed-data findings rest on small samples (the `choice` error subgroup is $n = 6$, and `noul`'s perfect accuracy on constructed data is 24 items). Old public bugs may appear in model training data, and the recent subsets used to check this are small (Mozilla 54, kernel 35). The protocol went through twelve revisions and the cross-model protocol three amendments, including a readout bug fixed after interim results were seen. On 3% of OLMo-2's requests most of the probability fell outside the allowed labels, and excluding them does not change its result. All revisions are dated and disclosed, and each was made before the data it affects existed. The number of revisions is nonetheless itself a limitation.

External validity. The study covers five models in one domain, software issue triage, across two workflows (Bugzilla bug triage and Kubernetes issue triage), with Jev at version 1.13.0. The general instruct models use our prompt and readout instead of a trained decision interface, and on Kubernetes they were below chance on the priority step. The Kubernetes study adds only one further workflow, and its development and test sets differ in period, format and label mix at once. Its leakage read was done by an AI assistant at the authors' direction, not by a human author as pre-registered. Its protocol has seven dated amendments, all written before any model output existed. Later model versions and other domains may behave differently.

XIV. CONCLUSION

We asked whether the uncertainty signals of a chained, typed decision workflow identify its failures, and whether they identify the failures worth human review. For the first question, the workflow's decisions predicted its failures better than its confidence did, on 500 new bugs and across five models, with failure measured against our constructed target. Most of the signal came from the action error rate. The result replicated on a separate Kubernetes workflow across a five-year gap, for Jev and three of the four other models.

These error rates depend on the label mix under which they were estimated, and they can invert when that mix changes. Re-weighting to a known deployment mix corrects them, but the label-free estimate of that mix helped only when the middle severity level dominated, and it reduced ranking quality on Kubernetes, where it did not.

The second question had a negative answer. Under the pre-registered costs, no review policy built from these signals beat no review. The expensive errors, wrongly accepted serious bugs, resemble correct accepts in both their action and their confidence, so neither signal isolates them. Review only pays off when a wrong accept costs much more than the review that would catch it.

XV. FUTURE SCOPE

The most pressing next step is to validate the target. In a pre-registered study, two human evaluators would independently pick the right action for a sample of held-out bugs. That would show how close our constructed target is to expert judgment, and whether the gaps come from the tracker labels, the action rule or real ambiguity.

A second need is a signal that separates expensive errors within an action. The oracle shows that such a signal would make review worthwhile, and OpenJev's weak within-action severity signal suggests where to look.

The setup should also be tried outside software triage, on newer model versions and against a supervised classifier as a baseline, and review value should be measured against real costs, with reviewers who can themselves be wrong.

Finally, we have pre-registered, but not yet run, a study on deterministic template content (3,103 units with value-neutral sentence frames and 13 automated audits). It tests the composition question under tighter control, comparing confidence-ranked against random overrides at a 10% budget scored by cost, and per-primitive against raw-product ranking. Its outcome does not affect the conclusions reported here.

Appendix A
Action rule and cost matrix

TABLE VIII
THE ACTION RULE $F(c_2, c_3)$. *Unclear* IS CHECKED FIRST.

$c_2 \setminus c_3$	0	1	2
1 (yes)	accept	defer	escalate
0 (no)	accept	accept	escalate
unclear	request evidence (any c_3)		

TABLE IX
FROZEN COST MATRIX $\mathrm{cost}(a, a^*)$. ONE UNIT OF HUMAN COST IS ADDED PER ESCALATION. "REQ. EV." = REQUEST EVIDENCE.

action \ target	accept	escalate	req. ev.	defer
accept	0	10	6	8
escalate	1	0	1	1
req. ev.	1	1	0	1
defer	1	1	1	0

Appendix B
Reproducibility

Every analysed Jev call used `jev-1.13.0`. Real-data calls: Mozilla 2,424 (including a label-wording arm) plus a 252-call smoke test; kernel 2,160 plus a 216-call smoke test; Kubernetes 9,738 (1,082 issues, three nodes, three repeats; the 270-call smoke test's issues were reused), with model version pinned after 9 smoke calls had resolved to `jev-1.14.0`; those calls and one version probe are kept and excluded. All analysed calls returned HTTP 200 and parsed. The four comparison models made 3,246 deterministic passes each on the Kubernetes issues, with no failures. Every dataset, raw response, execution manifest and analysis is hash-locked in the accompanying repository, as are the pre-registration documents and their revision history. The figure data in this paper are generated from committed result files by `make_figure_data.py`. An audit on 2026-10-23 found one defect: the held-out analysis seeded its per-tracker bootstrap with Python's salted string hash, so the HR2 one-sided lower bounds varied between runs (0.640–0.644 for the kernel, 0.759–0.763 for Mozilla); point estimates and test outcomes never changed. The seed is now deterministic (bounds 0.762 and 0.648), and every other analysis, re-run under a different hash seed, was byte-identical.

Appendix C
Data provenance and release

Sources. Bug reports come from the public Mozilla Bugzilla and Linux kernel Bugzilla, fetched through their REST interfaces. Texts were cleaned, privacy-scrubbed (names and e-mail addresses removed) and truncated to 2,000 characters; exclusion rules are applied by the dataset builders and recorded in manifests with SHA-256 hashes.

What is released. Bug identifiers, tracker URLs, reference labels, text hashes, every model output, every protocol with its commit history, all analysis code and every result file. Report texts are not redistributed: the builders regenerate them from the public trackers, and the hashes verify the regeneration. Bug-tracker links used in the split-integrity check are released as identifiers only.

What needs access. Re-running model calls needs a TypeSafe API key for Jev and a GPU for the open models (OpenJev at Hugging Face commit `b4a89f5` under vLLM 0.29.0; Qwen2.5, Phi-3.5 and OLMo-2 under `transformers`). Every table and figure in this paper can be regenerated offline from the released outputs, and a script checks every reported number against the result files.

Appendix D
Claims and the evidence behind them

Table X lists each main claim, whether it was confirmatory or exploratory, and the test behind it.

TABLE X
EACH MAIN CLAIM, ITS STATUS AND ITS TEST. CONFIRMATORY CLAIMS WERE SPECIFIED BEFORE THE DATA THEY USE EXISTED; EXPLORATORY CLAIMS WERE NOT, OR REUSE PREVIOUSLY ANALYSED BUGS.

Claim	Status	Test and outcome
Decision estimator ranks failures above final-step confidence (Jev, new bugs)	Confirmatory (HR1)	One-sided paired bootstrap, Holm; supported
Decision estimator above chance on each tracker	Confirmatory (HR2)	One-sided bootstrap, Holm; supported
Same contrast for four further models	Confirmatory, three amendments (X1, X2)	8 tests, Holm; all supported
Same contrast in an independent workflow (Kubernetes, Jev)	Confirmatory (K1, K2)	One-sided paired bootstrap, Holm; supported
Same contrast on Kubernetes for four further models	Confirmatory (XK1, XK2)	8 tests, Holm; 3 of 4 supported; post-hoc competence check: both competent models supported
Label-free correction transfers to Kubernetes	Confirmatory (K3)	Inconclusive, negative estimate
Estimator adds nothing beyond the action's error rate (Jev-style)	Exploratory	Paired bootstrap CI of AUROC difference
Error rates invert under label shift; re-weighting and BBSE repair the ranking (BBSE mix estimate biased)	Exploratory	AUROC with bootstrap CI on held-out bugs; synthetic known-shift test
Action rule beats random review (Jev, OpenJev)	Confirmatory (I1)	One-sided paired bootstrap, Holm; supported
Action rule beats confidence-ranked review	Confirmatory (I2)	Not supported for any model
No 10% review policy beats no review	Descriptive, pre-registered design	Point estimates under the frozen cost table
Whether review pays depends on the cost ratio	Exploratory (post hoc, planned)	Frozen rankings re-scored on a cost grid
Case-level score not viable	Pre-registered feasibility check	Simulation on development data; stop rule applied
Primitive confidence profiles, threshold transfer, severity compression	Exploratory / descriptive	Development bugs, bootstrap and Wilson intervals

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Following IEEE policy on AI-generated content, we disclose that this work used Claude (Opus 5.5), a large language model by Anthropic [47], accessed through Claude Code (https://claude.com/claude-code). Claude wrote the analysis code, including the code that produces the figures and tables, and ran the analyses under our direction. The AI-assisted steps described in the text, the Kubernetes leakage read and the target-validity pilot, also used Claude. Claude prepared early drafts of the text, which we then edited and reviewed manually in full. For this version, Claude was used only for light rephrasing of our edited text. We directed the study, approved every pre-registration and model run, checked every reported number against the committed result files, and take full responsibility for the content.

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